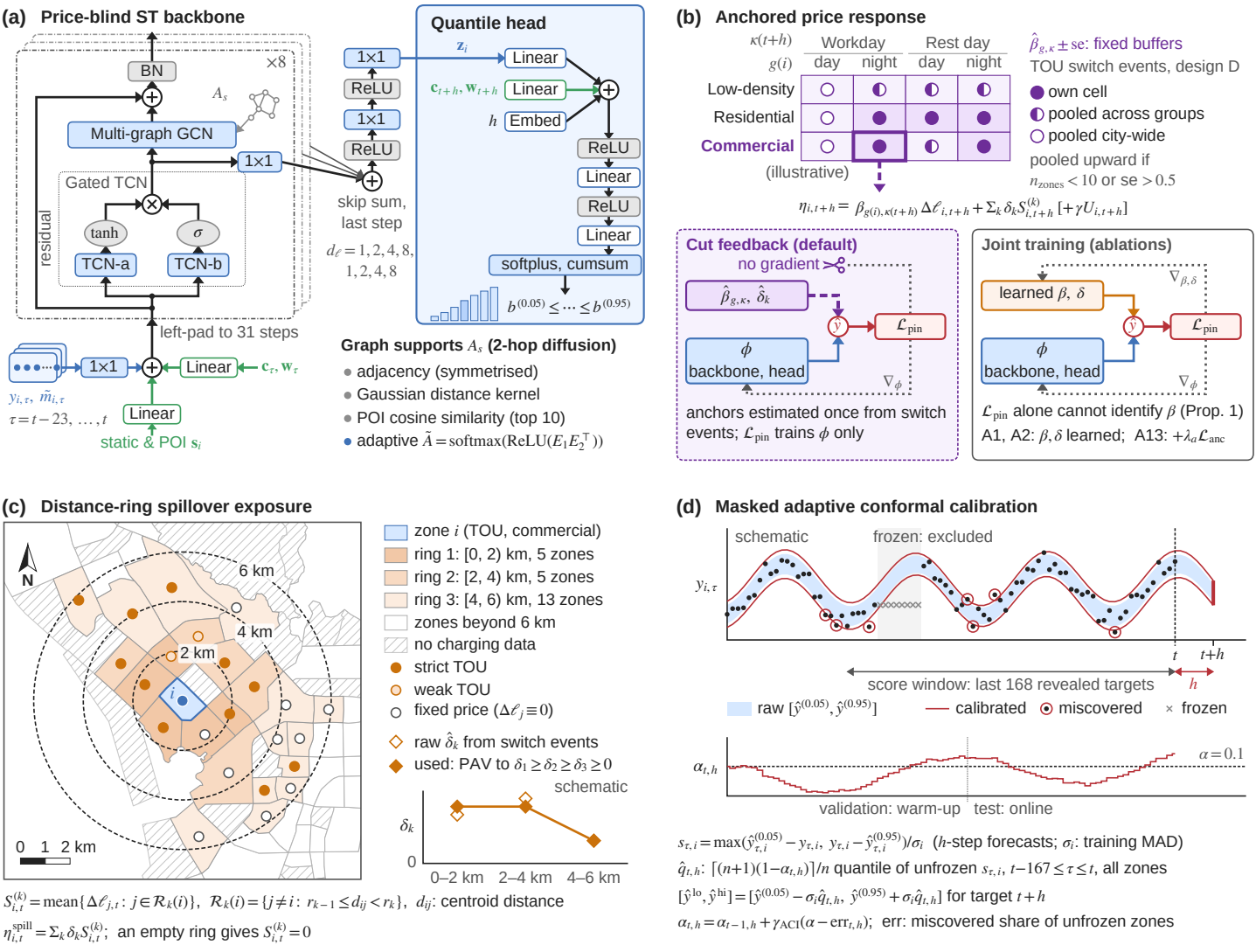


$\eta_{i,t+h} = \beta_{g(i),\kappa(t+h)}\,\Delta\ell_{i,t+h} + \Sigma_k \delta_k S^{(k)}_{i,t+h}\,[+\gamma U_{i,t+h}]$

$S^{(k)}_{i,t} = \text{mean}\{\Delta\ell_{j,t} : j \in \mathcal{R}_k(i)\}$, $\mathcal{R}_k(i) = \{j \neq i : r_{k-1} \le d_{ij} < r_k\}$, d_{ij}: centroid distance

$\eta^{\text{spill}}_{i,t} = \Sigma_k \delta_k S^{(k)}_{i,t}$; an empty ring gives $S^{(k)}_{i,t} = 0$

$s_{\tau,i} = \max(\hat{y}^{(0.05)}_{\tau,i} - y_{\tau,i},\ y_{\tau,i} - \hat{y}^{(0.95)}_{\tau,i})/\sigma_i$ (h-step forecasts; σ_i: training MAD)

$\hat{q}_{t,h}$: $\lceil (n+1)(1-\alpha_{t,h}) \rceil / n$ quantile of unfrozen $s_{\tau,i}$, $t-167 \le \tau \le t$, all zones

$[\hat{y}^{\text{lo}}, \hat{y}^{\text{hi}}] = [\hat{y}^{(0.05)} - \sigma_i \hat{q}_{t,h},\ \hat{y}^{(0.95)} + \sigma_i \hat{q}_{t,h}]$ for target $t+h$

$\alpha_{t,h} = \alpha_{t-1,h} + \gamma_{\text{ACI}}(\alpha - \text{err}_{t,h})$; err: miscovered share of unfrozen zones